Case Vignettes: Thyroid, Pituitary and Parathyroid

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- No conflicts of interest to disclose

Learning Objectives

By exam time, you should be able to do the following:

- Summarize how to interpret laboratory tests to diagnose thyroid dysfunction.
- Understand the management and treatment of central hypothyroidism and central hyperthyroidism.
- Summarize the laboratory evaluation and diagnostic approach to pituitary adenomas.
- Explain the treatment options and management of acromegaly, prolactinoma, and Cushing disease.
- Understand the pathophysiology of hypercalcemia
- Identify and understand PTH-mediated and non-PTH mediated hypercalcemia.

Thyroid

Review: Guess the Type of Thyroid Dysfunction!

Choices:

Primary hypothyroidism

Subclinical hypothyroidism

Central hypothyroidism

Primary hyperthyroidism

Subclinical hyperthyroidism

Central hyperthyroidism

TSH	Free T4	T3	Diagnosis
↓	↑	↑	
↑/Normal	↑	↑	
↓	Normal	Normal	
↑	↓	↓	
↓/Normal	↓	↓	
↑	Normal	Normal	

Case 1

- A 45-year-old woman presents with fatigue, weight gain, and cold intolerance. Menstrual periods have been irregular.

- She reports progressive headaches and occasional blurred vision over the past 6 months.

Physical Examination

- Dry skin, periorbital puffiness
- Mild bitemporal hemianopia
- No thyroid enlargement

Laboratory Findings

- TSH: 0.8 mIU/L (normal 0.5–5.0)
- Free T4: 0.5 ng/dL (low)
- Cortisol: 7 µg/dL (low-normal)
- Prolactin: mildly elevated
- Na 132 (mildly low)
- K 4.5 (normal)

What type of thyroid dysfunction is this?

Imaging

- MRI of the brain shows a 2.5 cm pituitary macroadenoma compressing the optic chiasm.



Diagnosis

- Central Hypothyroidism due to Pituitary Macroadenoma
- Additional labs show:
  - Normal IGF-1, FSH and LH

Which if the following best explains why TSH is not reliable for monitoring treatment in central hypothyroidism?

A. TSH is suppressed by levothyroxine therapy

B. TSH secretion is impaired due to pituitary dysfunction

C. TSH assay interference is common in pituitary disease

D. TSH is secreted in a pulsatile manner

Before starting levothyroxine in patient with central hypothyroidism, what is important to always do?

A. Check serum calcium
B. Exclude adrenal insufficiency
C. Perform thyroid ultrasound
D. Start growth hormone replacement

Management

- Start levothyroxine replacement (guided by free T4, not TSH)
- Assess for adrenal insufficiency before thyroid replacement
  - Levothyroxine increases the metabolic rate/turnover of cortisol
- Refer to neurosurgery for consideration of tumor resection

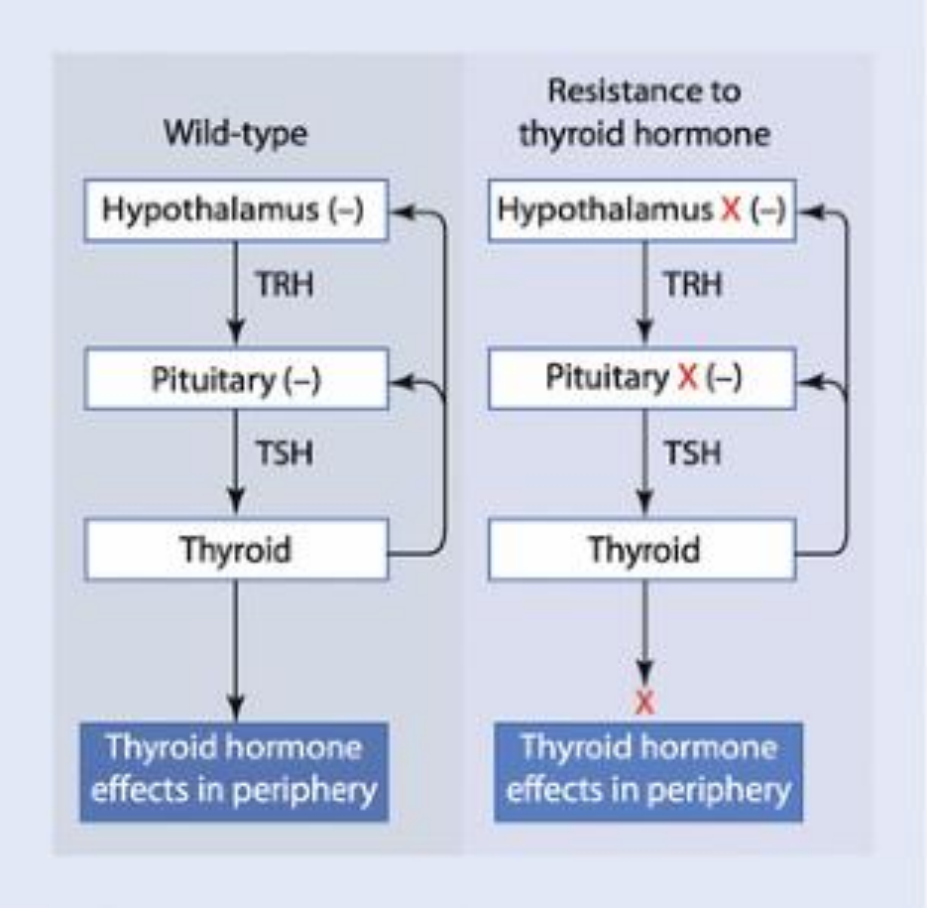
- She passed a cosyntropin stimulation test
  - Cortisol >14-16 one hour after 250mcg cosyntropin
- Her symptoms resolved after 6-8 weeks on levothyroxine therapy

Case 2

- The patient is a 58-year-old female with a history of a multinodular goiter who presents with the chief complaint of weight loss and occasional palpitations. She has unintentionally lost 8 pounds over the past year and feels that her goiter is getting a bit bigger.
- She has had a fine needle aspiration biopsy of her nodules and they are benign.
- She denies any other complaints
- On exam her pulse is 98, BP 120/74. She has a “lumpy” goiter on exam and a fine tremor in her hands with out-stretched arms.

- Labs show a TSH of 5.4 mIU/L (normal 0.45-4.5) with an elevated free t4 and free t3.
- What type of thyroid dysfunction is this?

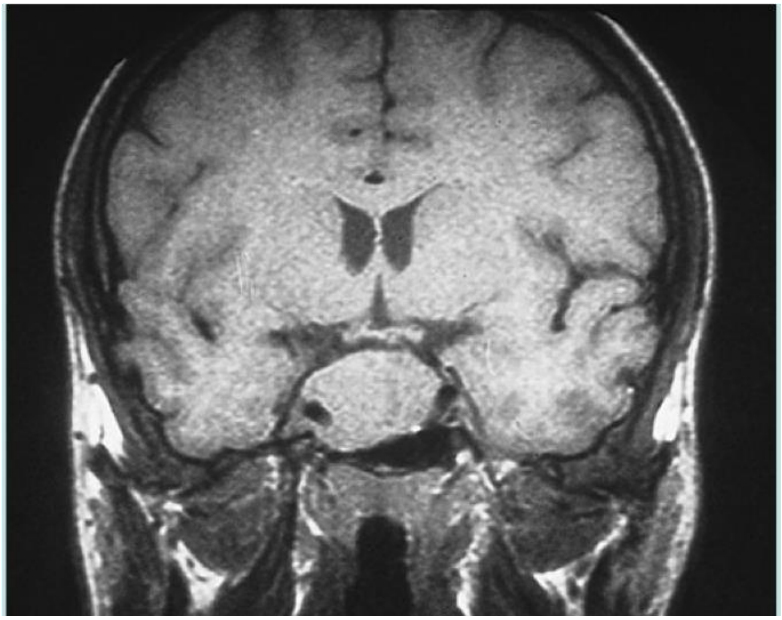
Resistance to thyroid hormone (RTH)



Resistance to thyroid hormone (RTH)

- Resistance to thyroid hormone (RTH) is an inherited syndrome characterized by reduced responsiveness of target tissues to thyroid hormone.
- The disorder is characterized by high serum concentrations of free T4 and usually free T3, accompanied by normal or slightly high serum thyroid stimulating hormone (TSH) concentrations.
- Most patients with RTH is the paucity of symptoms and signs of thyroid dysfunction despite the presence of high serum T4 and T3 concentrations.
- Often presents with a goiter due to thyroid over-production
- Does not require treatment (sometimes beta-blocker)

TSH Producing Adenoma (very rare!)



- The glycoprotein hormone family (TSH, LH, FSH,HCG) consist of two peptides –alpha and beta.
- LH, FSH, TSH, HCG share the same common alpha subunit and have unique beta subunits.
- Can measure alpha subunit to see if elevated in suspected TSH producing adenoma.

TSH Producing Adenoma

- What agent could you use to treat her medically?
  - **Somatostatin medications can restore euthyroidism prior to surgery**
  - **Beta-blockers to treat acute symptoms**
  - **Antithyroid therapy of any type, radioiodine or drug (eg, thionamide), is not indicated for the treatment of patients with TSH-secreting adenomas, because sustained reductions in thyroid hormone secretion would be expected to increase TSH secretion and stimulate tumor growth.**

Pituitary

Case 1

- A 60-year-old male with a history of prediabetes, HTN, colonic polyps, sleep apnea and osteoarthritis presents to your office because his new dentist was concerned about his somewhat enlarged tongue and dental spacing.
- The patient and his wife haven’t noticed much change in his appearance.
- On physical exam his BP and pulse are normal. He has skin tags on neck and upper back.





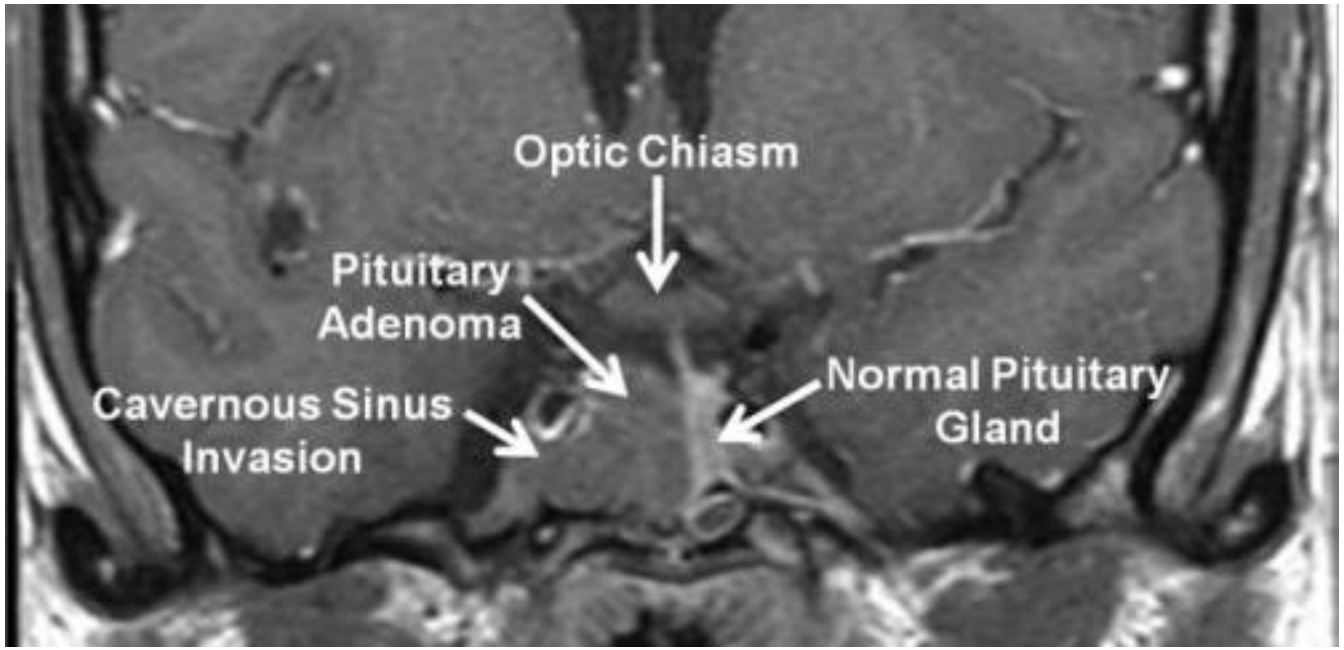
Since you had excellent training in medical school, what test should you order to make the correct diagnosis?

A. IGF-1
B. Growth hormone
C. AM Cortisol
D. TRH
E. Prolactin

- Unlike growth hormone (GH), serum IGF-1 concentrations do not vary from hour to hour according to food intake, exercise, or sleep, but instead they reflect integrated GH secretion during the preceding day or longer.
- Serum IGF-1 concentrations are elevated in virtually all patients with acromegaly and provide excellent discrimination from normal individuals.

- IGF-1 is 540 ng/mL (normal ~100-300) – this is a screening test
- Confirmatory test for acromegaly 75-gram oral glucose tolerance test
- His 2-hour GH level is 3.8 ng/mL confirming the diagnosis of acromegaly (>0.4 is abnormal)
- What should we do next?

MRI coronal view



2cm pituitary adenoma

Tough diagnosis to make early

- The average interval from the onset of symptoms until diagnosis is approximately 12 years
- At diagnosis, approximately 75 percent of patients have macroadenomas (tumor diameter 10 mm or greater), and some of the adenomas extend to the parasellar or suprasellar regions



May 1998



June 1999



December 2001



June 2003



June 2005



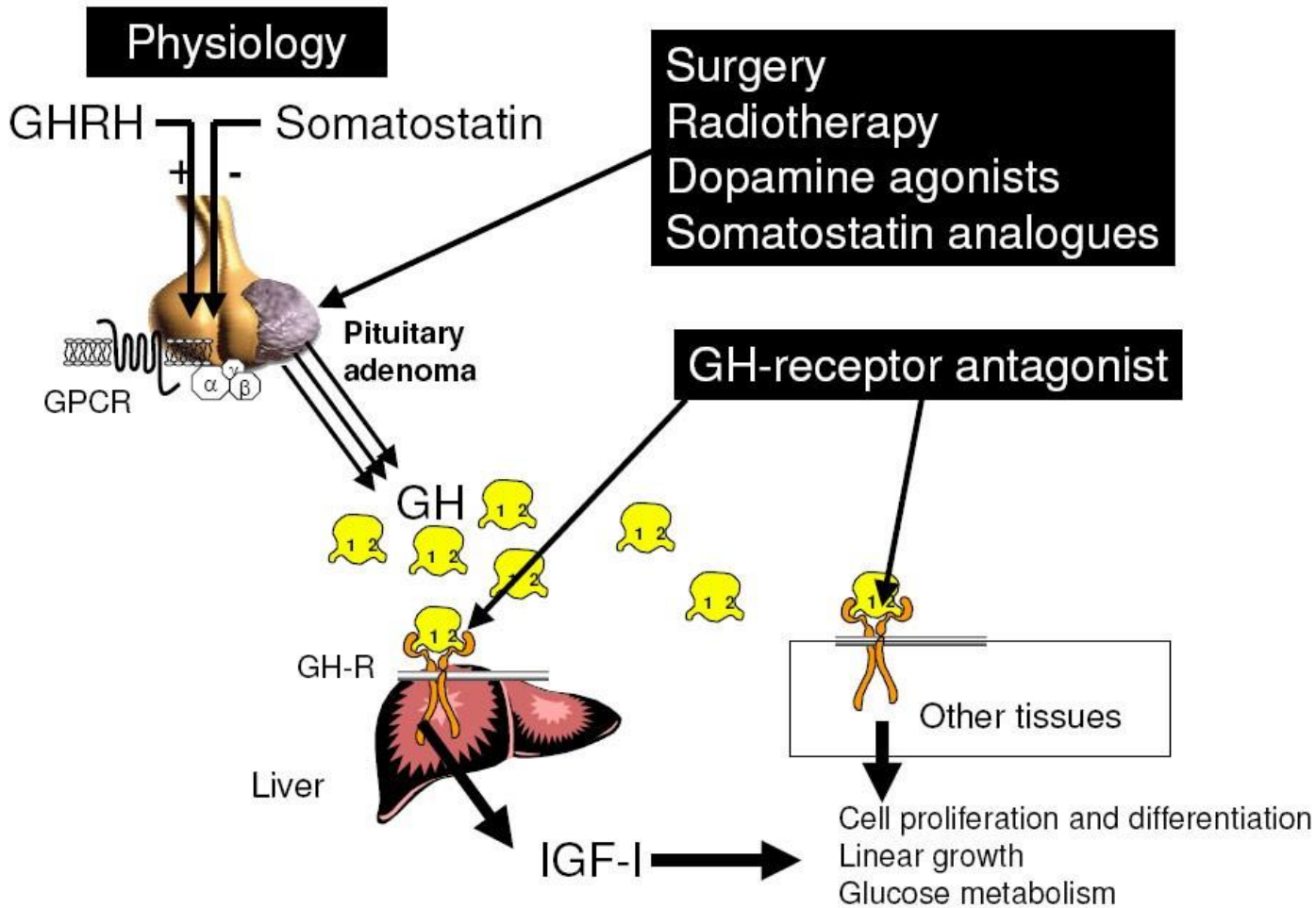
June 2006
(15days before diagnosed)



Back to our case

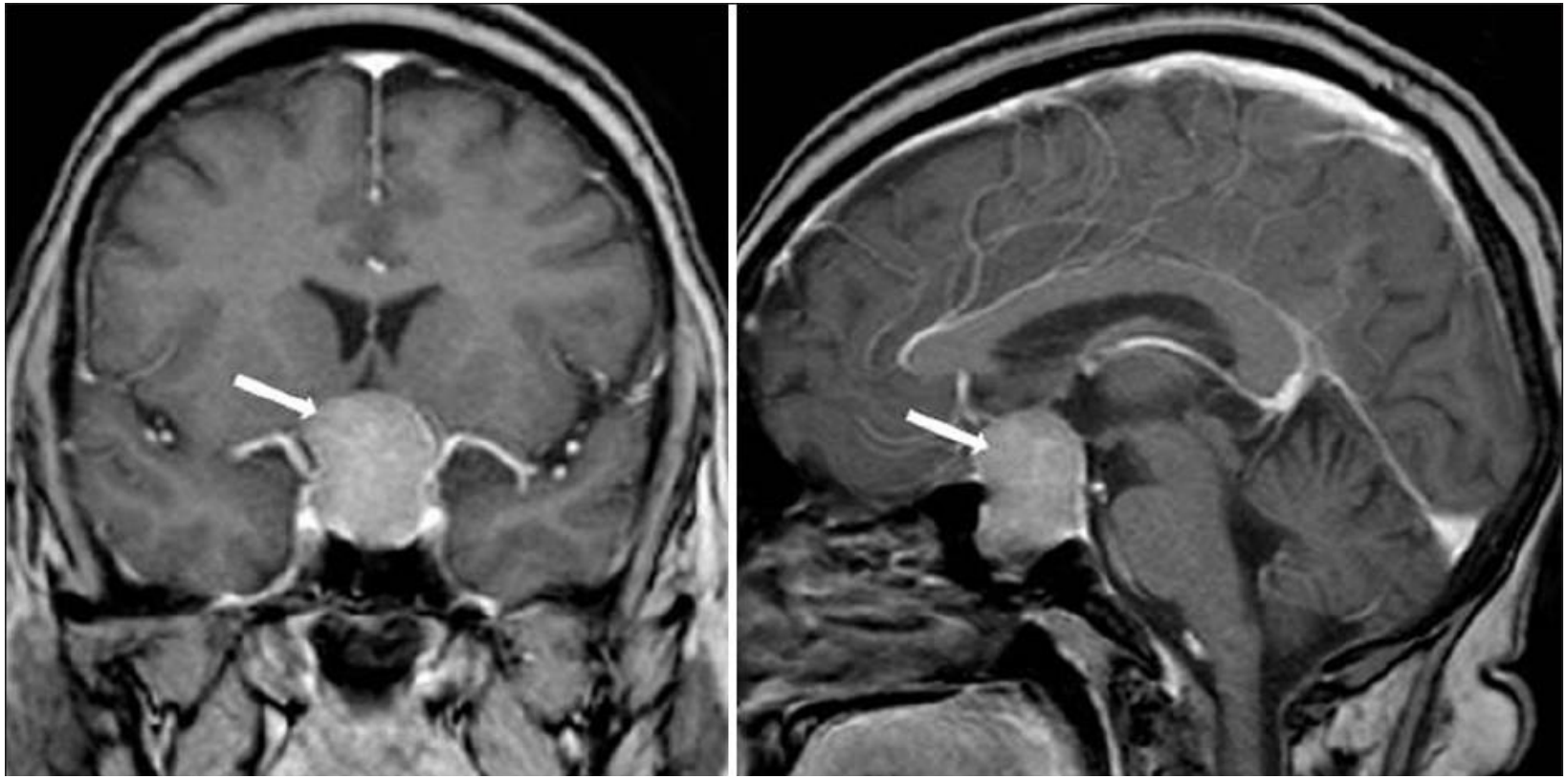
The patient has a transsphenoidal resection by neurosurgery. He does well post operatively. Six months later, his IGF-1 is still elevated. MRI does not show any residual tumor. The next best step in his management is:

A. Exploratory pituitary surgery
B. Somatostatin analog
C. Observation
D. Ketoconazole


Physiology
GHRH
Somatostatin
+
-
Surgery
Radiotherapy
Dopamine agonists
Somatostatin analogues
Pituitary adenoma
GPCR
α
β
γ
GH-receptor antagonist
GH
GH-R
Liver
Other tissues
IGF-I
Cell proliferation and differentiation
Linear growth
Glucose metabolism

Case 2

- The patient is a 50-year-old male who presents with bitemporal hemianopsia and left sided ptosis. He denies any other symptoms other than occasional headache.
- He has a history of decreased libido and hypogonadism (low testosterone).
- MRI shows a 3.5 cm pituitary mass



What labs should we order?

- With macroadenomas >10mm (1cm), need to assess for hormonal deficiencies in addition to hormonal hypersecretion
- Microadenomas <10mm (1cm), only hypersecretion is a concern

Labs

- Prolactin 3688 ng/mL (4-15.2 ng/mL)
- Testosterone 46 ng/dL (264-916)
- LH low
- FSH low
- Cortisol (8AM) 14 ug/dL (6-19)
- 24-hr urine free cortisol normal
- ACTH normal
- IGF-1 normal

What is the most appropriate initial treatment?

- A. Transsphenoidal resection of the adenoma
- B. Somatostatin analogue
- C. XRT to the sella
- D. Dopamine agonist

Prolactinoma

- Most common hypersecreting pituitary adenoma
- Women often diagnosed earlier due to oligomenorrhea or amenorrhea
- Men more likely than women to have macroprolactinoma
- Dopamine agonists are the mainstay of treatment; surgery is used in cases of immediate danger due to optic chiasm compression, medication failure/intolerance, and select small tumors.
- Cabergoline (twice a week) or bromocriptine (daily)
- Dopamine agonists decrease the size of adenomas in most cases

Case 3

- The patient is a 42 year old female with PMH of PCOS, type 2 diabetes, HTN and obesity.
- Takes Metformin twice a day and semaglutide injections weekly, Amlodipine and losartan. She is not on any glucocorticoids.
- Has been gaining weight for many years and c/o easy bruising.
- BP 148/86, pulse 80, BMI 48

Physical Exam

- Moon facies
- Proximal muscle weakness
- Abdominal striae
- Bruises on arms and legs
- Mild facial plethora



Labs

- 24-hr UFC 150 mcg/24hr (4-40 mcg/24hr)
- 1mg dexamethasone suppression 6.2 ug/dL (<1.8 ug/dL)
- ACTH 80 pmol/L (10-60 pmol/L)
- Na 142, K 3.3
- 8AM cortisol 14 mcg/dL (8-10AM: 6-18 mcg/dL)

- MRI 7mm pituitary adenoma

What is the recommended initial treatment for this condition?

- A. Transsphenoidal surgery
- B. Mifepristone
- C. Ketoconazole
- D. XRT to the sella

Parathyroid

Case 1

A 59 yr old female without significant past medical history presents for annual wellness exam. She had laboratory studies done and was told that her calcium was high. She is unaware of any prior abnormal labs. She feels well and denies any bone pain, fractures, or gastrointestinal symptoms.

Medications: none

Social history: smoked 1ppd for 10 years, quit at age 35. Alcohol use: 1-2 drinks per week (wine). Rare exercise

Case 1

Family history is negative for kidney stones or calcium issues

Physical Examination:

VSS stable, no evidence of dehydration, well appearing
HEENT exam: wnl, no palpable thyroid mass or nodules
CV: RRR
Pulm: CTA bilaterally
GI: normal BS/ non-tender
Extr: warm, no edema

Case 1

Labs:

iPTH: 59 pg/mL (10-65)

Calcium: 10.8mg/dL (8.1- 10.3)

Albumin: 4.0g/L (normal range)

1,25-OH vitamin D : 50pg/mL (16-65)

25-OH vitamin D : 15ng/mL (20-75)

Creatinine: 1.0mg/dL (normal range)

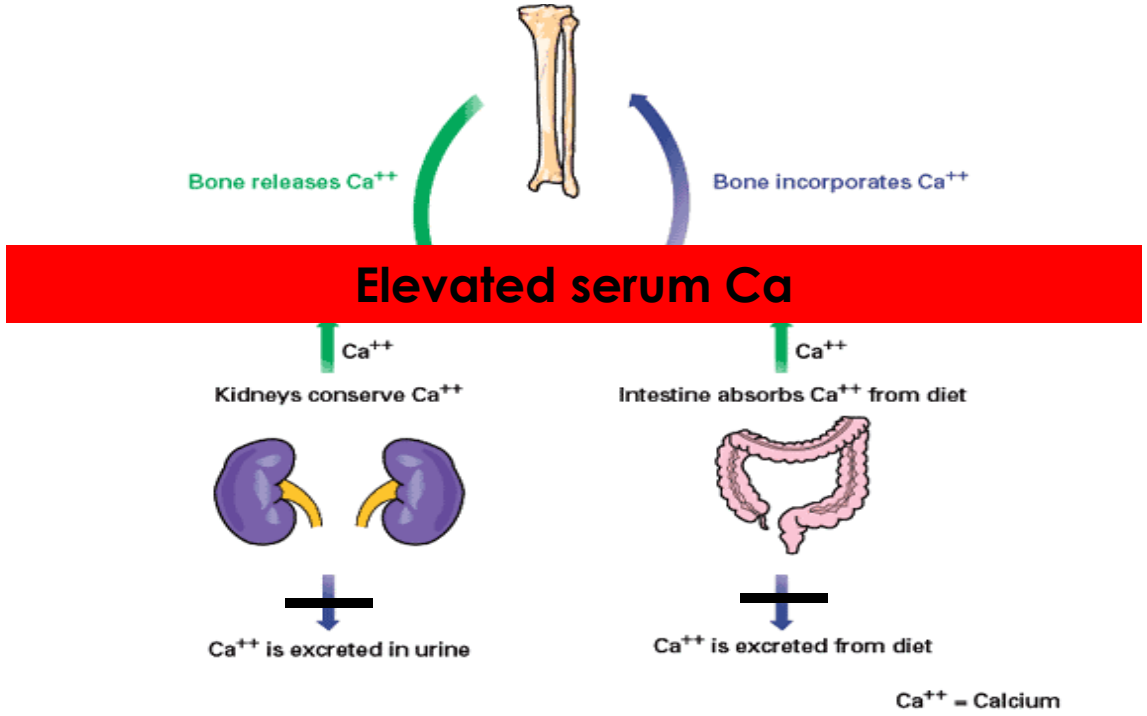
Urinary calcium excretion: 230 mg/24 hr (normal range)

Which of the following is the most likely diagnosis?

A. PTHrP (PTH related peptide) mediated hypercalcemia
B. Primary hyperparathyroidism
C. Sarcoidosis
D. Secondary hyperparathyroidism

Pathophysiology of Hypercalcemia

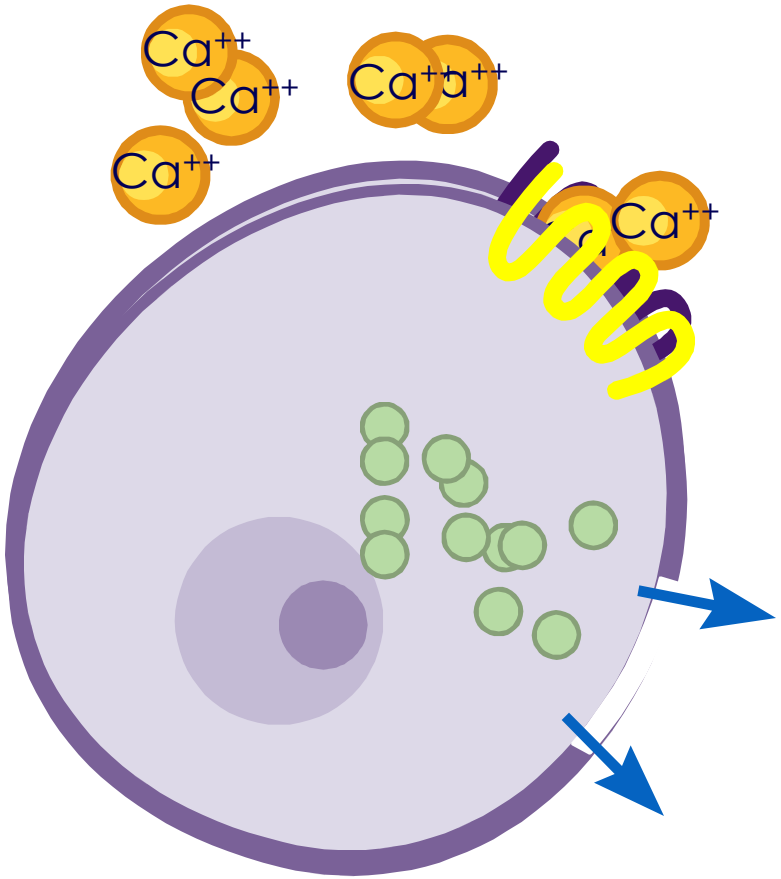
1. Increased bone resorption (turnover)



3. Increased renal reabsorption

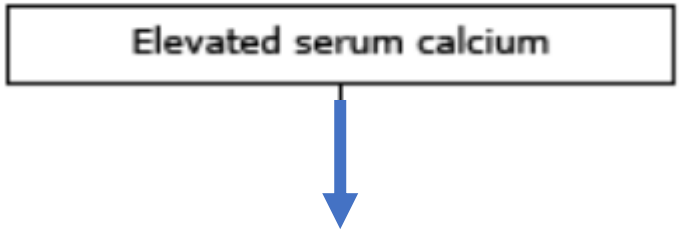
2. Increased Gut absorption

Normal Physiology:
The Parathyroid Cell Calcium Receptor (CaR)



- The CaR is the principal regulator of PTH secretion
- Hypocalcemia → ↑ PTH secretion
- Hypercalcemia – ↓ PTH secretion

Algorithmic Approach to Hypercalcemia



What next?

A. Verify calcium (Repeat calcium with albumin)

B. Check PTH – is this PTH mediated ?

Causes of Hypercalcemia

PTH Dependent:
Primary Hyperparathyroidism
Tertiary Hyperparathyroidism
Parathyroid carcinoma
Familial hypocalciuric hypercalcemia

PTH Independent:
HyperCa of malignancy
Granulomatous disease
Vit D / Vit A intoxication
Milk-alkali Syndrome
Multiple Myeloma
Medications (lithium)
Thyrotoxicosis

Primary Hyperparathyroidism

A common disorder – mostly found in the **outpatient** setting

- Symptoms *may* include: polyuria, polydipsia, nephrolithiasis, bone pain/fracture, fatigue, weakness, depression

Single adenoma in 80%; 4 gland hyperplasia in 15%; multiple adenomas in 5%

- Most are sporadic
- Familial forms-MEN 1 or MEN 2A; usually hyperplasia
- Parathyroid carcinoma-rare

Why do we treat?

- Osteoporosis, Renal disease/ Stones, Quality of life

Primary Hyperparathyroidism Laboratory Findings

Hypercalcemia is always present

PTH is either elevated or **inappropriately normal**

Phos ↓ or normal

Urinary calcium may be elevated

Case 1 Continued

Labs:
Calcium: 10.8 (8.1- 10.3)
Urinary calcium excretion: 230 (normal range)

You have discussed the diagnosis of primary hyperparathyroidism with the patient. You order a couple of more tests:

Dexa Scan shows osteoporosis at the left hip and osteopenia at the lumbar spine

X-ray of abdomen- no evidence of calcium containing renal stones

What is the next best step?

A. Advise her to have surgery (parathyroidectomy) because she is at high risk for kidney stones
B. Advise her to have surgery (parathyroidectomy) based on her dexa scan
C. Advise her to have surgery (parathyroidectomy) based on her serum calcium level
D. Advise her that she should not have surgery (parathyroidectomy) based on her age of 59
E. None of the above

Primary Hyperparathyroidism: Evaluation and Treatment

- Labs to diagnose
- Check for complications/ surgical indicators
  - DXA
  - X-ray or other imaging for kidney stones
  - Urine calcium

- Imaging to look for parathyroids?
  - Do this if planning surgery **(Not for diagnosis)**
    - Sestamibi Scan
    - Neck ultrasound

Indications for Parathyroidectomy

Category	Indication
Age	< 50 years
Serum Calcium	> 1.0 mg/dL (0.25 mmol/L) above ULN
Skeletal	T-score ≤ –2.5 at lumbar spine, total hip, femoral neck, or distal 1/3 radius Vertebral fracture on imaging
Renal Function	eGFR or creatinine clearance < 60 mL/min
Nephrolithiasis / Nephrocalcinosis	Kidney stones or nephrocalcinosis on imaging
Urinary Calcium Excretion	Women: > 250 mg/day Men: > 300 mg/day
Patient Preference	Reasonable if patient desires definitive therapy and surgical risk acceptable

Treatment

- Parathyroidectomy- Surgery
  - Mostly minimally invasive
  - Sometimes must perform exploratory surgery
  - Experienced surgeon
  - 95% cure rate

- Medical therapy
  - Avoid dehydration and meds which may worsen hypercalcemia
  - Optimize bone health
  - Cinacalcet (Sensipar) increases the sensitivity of CASR to serum calcium (to help lower serum calcium)

Case 2

A 72-year-old man with a history of smoking presents to the ER because of cough, shortness of breath, anorexia, and weight loss for the past few months. Physical examination shows a cachectic and lethargic man in mild respiratory distress, and rhonchi in the right upper lobe.

Labs show a serum calcium of 13 mg/dl (N:8.1-10.3), serum creatinine if 1.4mg/dL (range 0.7- 1.4). A chest X-ray shows a 6-cm right hilar mass.

Which of the following can you assume to be correct?

A. Hypercalcemia is due to production of 1,25-OH vitamin D by the lung tumor
B. PTH levels should be suppressed
C. Recommended and definitive treatment will be parathyroidectomy
D. The lung tumor is most likely small cell carcinoma

Case 2 continued

The serum calcium is 13 mg/dl (N:8.1-10.3), serum creatinine is 1.4mg/dL

How would treat/ manage this patient's hypercalcemia?

Management & Treatment of Severe Hypercalcemia

#1. <u>Hydration</u> (IVF, Normal saline)

Then, consider underlying cause and treat:

-Calcitonin, sub-cutaneous injection (can work within hours, though not as powerful as bisphosphonates), prevents calcium release from bone and reduces reabsorption in the kidneys

-Bisphosphonates: IV Zolendronic acid (may take several days to work)- Block calcium release from bones

-Glucocorticoids: Inhibit Gut absorption of Calcium: (ex hypercalcemia due to sarcoidosis or lymphoma – 1,25 D mediated)

-Other treatments

- Denosumab- also used for osteoporosis, reduces bone resorption, in cases refractory to bisphosphonates or if eGFR is too low to use bisphosphonates)
- Hemodialysis

THANK YOU!
Q+A